

explain analyze select id, g from students where g > 8 and g < 95 order by g desc;

```
Gather Merge (cost=3918953.46..8933593.35 rows=43856588 width=8) (actual time=20160.191..30948.339 rows=43800301.00 loops=1)
  Workers Planned: 2
  Buffers: shared hit=789 read=952650, temp read=189384 written=189906
    → Sort (cost=3917953.43..3962893.96 rows=17940212 width=8) (actual time=20011.430..22019.843 rows=14333433.67 loops=3)
      Sort Key: g DESC
      Sort Method: external merge Disk: 25376848
      Buffers: shared hit=789 read=952650, temp read=189384 written=189906
      Worker 0: Sort Method: external merge Disk: 25130448
      Worker 1: Sort Method: external merge Disk: 25254048
      → Parallel Seq Scan on students (cost=0.00..1265901.48 rows=17940212 width=8) (actual time=128.033..13808.115 rows=14333433.67 loops=3)
        Filter: ((g > 8) AND (g < 95))
        Rows Removed by Filter: 233233
        Buffers: shared hit=715 read=952650
Planning Time: 0.075 ms
JIT:
  Functions: 12
  Options: Inlining true, Optimization true, Expressions true, Deforming true
  Timing: Generation 0.942 ms (Deform 0.309 ms), Inlining 259.131 ms, Optimization 67.196 ms, Emission 55.005 ms, Total 382.274 ms
Execution Time: 33167.397 ms

Its for non indexed, and 50M rows
```

explain analyze select id, g from students where g > 8 and g < 95 order by g desc limit 1000;

```
Limit (cost=2250543.97..2250660.40 rows=1000 width=8) (actual time=16603.312..16609.001 rows=1000.00 loops=1)
  Buffers: shared hit=1071 read=952368
    → Gather Merge (cost=2250543.97..2265183.98 rows=43856589 width=8) (actual time=16574.034..16579.708 rows=1000.00 loops=1)
      Workers Planned: 2
      Workers Launched: 2
      Buffers: shared hit=1071 read=952368
        → Sort (cost=2209543.95..2209394.48 rows=17940212 width=8) (actual time=16557.726..16557.813 rows=1000.00 loops=1)
          Sort Key: g DESC
          Sort Method: top-N heapsort Memory: 11548
          Buffers: shared hit=1071 read=952368
          Worker 0: Sort Method: top-N heapsort Memory: 11548
          Worker 1: Sort Method: top-N heapsort Memory: 11548
          → Parallel Seq Scan on students (cost=0.00..1265901.48 rows=17940212 width=8) (actual time=81.805..13611.605 rows=14333433.67 loops=3)
            Filter: ((g > 8) AND (g < 95))
            Rows Removed by Filter: 233233
            Buffers: shared hit=997 read=952368
Planning Time: 0.078 ms
JIT:
  Functions: 13
  Options: Inlining true, Optimization true, Expressions true, Deforming true
  Timing: Generation 1.066 ms (Deform 0.353 ms), Inlining 181.089 ms, Optimization 52.462 ms, Emission 37.018 ms, Total 271.635 ms
Execution Time: 16609.497 ms
```

Afdter Indexing g

```
Gather Merge (cost=7918593.59..8932648.23 rows=43851483 width=8) (actual time=23407.568..34209.309 rows=43800301.00 loops=1)
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=1542 read=951897, temp read=189383 written=189904
    → Sort (cost=7917593.56..3962438.86 rows=17938118 width=8) (actual time=23298.711..25307.638 rows=14333433.67 loops=5)
      Sort Key: g DESC
      Sort Method: external merge Disk: 25248048
      Buffers: shared hit=1542 read=951897, temp read=189383 written=189904
      Worker 0: Sort Method: external merge Disk: 20938448
      Worker 1: Sort Method: external merge Disk: 25574048
      → Parallel Seq Scan on students (cost=0.00..1265865.00 rows=17938118 width=8) (actual time=113.332..17011.904 rows=14333433.67 loops=5)
        Filter: ((g > 8) AND (g < 95))
        Rows Removed by Filter: 233233
        Buffers: shared hit=1668 read=951897
Planning:
  Buffers: shared hit=27 read=9 dirtied=2
Planning Time: 12.182 ms
JIT:
  Functions: 12
  Options: Inlining true, Optimization true, Expressions true, Deforming true
  Timing: Generation 0.960 ms (Deform 0.285 ms), Inlining 218.651 ms, Optimization 68.137 ms, Emission 52.198 ms, Total 339.946 ms
Execution Time: 36379.358 ms
```

explain analyze select id, g from students where g > 8 and g < 95 order by g desc limit 1000;

```
Limit (cost=0.56..1933.77 rows=1000 width=8) (actual time=2.282..3.871 rows=1000.00 loops=1)
  Buffers: shared hit=768 read=0
    → Index Scan Backward using g_index on students (cost=0.56..83227396.87 rows=43851483 width=8) (actual time=2.281..3.683 rows=1000.00 loops=1)
      Index Cond: ((g > 8) AND (g < 95))
      Index Searches: 1
      Buffers: shared hit=768 read=0
Planning:
  Buffers: shared hit=20
Planning Time: 0.190 ms
Execution Time: 4.001 ms
(10 rows)
```

explain analyze select id, g from students where g > 10 and g < 20 order by g desc limit 1000;

```
Limit (cost=0.56..2088.01 rows=1000 width=8) (actual time=2.221..3.943 rows=1000.00 loops=1)
  Buffers: shared hit=752 read=21
    → Index Scan Backward using g_index on students (cost=0.56..8762738.72 rows=4197833 width=8) (actual time=2.219..3.800 rows=1000.00 loops=1)
      Index Cond: ((g > 10) AND (g < 20))
      Index Searches: 1
      Buffers: shared hit=752 read=21
Planning:
  Buffers: shared hit=20
Planning Time: 0.131 ms
Execution Time: 4.074 ms
(10 rows)
```

explain analyze select id, g from students where g > 8 and g < 95 order by g desc;

```
Gather Merge (cost=1472145.96..1961052.79 rows=4197833 width=8) (actual time=16948.944..18078.041 rows=4098746.00 loops=1)
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=2161 read=951278, temp read=19848 written=19901
    → Sort (cost=1471145.93..1475518.67 rows=1749097 width=8) (actual time=16914.167..17107.819 rows=1499582.00 loops=3)
      Sort Key: g DESC
      Sort Method: external merge Disk: 2687248
      Buffers: shared hit=2161 read=951278, temp read=19848 written=19901
      Worker 0: Sort Method: external merge Disk: 2627248
      Worker 1: Sort Method: external merge Disk: 2639248
      → Parallel Seq Scan on students (cost=0.00..1265865.00 rows=1749097 width=8) (actual time=89.072..16318.163 rows=1499582.00 loops=3)
        Filter: ((g > 10) AND (g < 20))
        Rows Removed by Filter: 15167085
        Buffers: shared hit=2087 read=951278
Planning:
  Buffers: shared hit=20
Planning Time: 0.126 ms
JIT:
  Functions: 12
  Options: Inlining true, Optimization true, Expressions true, Deforming true
  Timing: Generation 0.937 ms (Deform 0.299 ms), Inlining 158.523 ms, Optimization 61.520 ms, Emission 44.615 ms, Total 265.595 ms
Execution Time: 18310.276 ms
(22 rows)
```

explain (analyze, buffers) select id, g from students where g > 10 and g < 20 order by g desc

```
Gather Merge (cost=1472145.96..1961052.79 rows=4197833 width=8) (actual time=16135.685..17046.621 rows=4098746.00 loops=1)
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=2013 read=950526, temp read=19848 written=19901
    → Sort (cost=1471145.93..1475518.67 rows=1749097 width=8) (actual time=16103.397..16260.068 rows=1499582.00 loops=3)
      Sort Key: g DESC
      Sort Method: external merge Disk: 2678448
      Buffers: shared hit=2013 read=950526, temp read=19848 written=19901
      Worker 0: Sort Method: external merge Disk: 2614048
      Worker 1: Sort Method: external merge Disk: 2660848
      → Parallel Seq Scan on students (cost=0.00..1265865.00 rows=1749097 width=8) (actual time=84.993..15531.587 rows=1499582.00 loops=3)
        Filter: ((g > 10) AND (g < 20))
        Rows Removed by Filter: 15167085
        Buffers: shared hit=2039 read=950526
Planning:
  Buffers: shared hit=32 read=8 dirtied=3
Planning Time: 12.894 ms
JIT:
  Functions: 12
  Options: Inlining true, Optimization true, Expressions true, Deforming true
  Timing: Generation 0.864 ms (Deform 0.264 ms), Inlining 150.266 ms, Optimization 56.826 ms, Emission 46.839 ms, Total 254.794 ms
Execution Time: 17243.445 ms
```

explain (analyze, buffers) select id, g from students where g > 10 and g < 20 order by g desc limit 1000;

```
Limit (cost=0.56..2246.22 rows=1000 width=8) (actual time=2.121..3.203 rows=1000.00 loops=1)
  Buffers: shared hit=771 read=5
    → Index Only Scan Backward using g_index on students (cost=0.56..9426883.05 rows=4197833 width=8) (actual time=2.120..3.139 rows=1000.00 loops=1)
      Index Cond: ((g > 10) AND (g < 20))
      Heap Fetches: 1000
      Index Searches: 1
      Buffers: shared hit=771 read=5
Planning:
  Buffers: shared hit=20
Planning Time: 0.162 ms
Execution Time: 3.257 ms
(11 rows)
```

After Vacuum

explain (analyze, buffers) select id, g from students where g > 10 and g < 20 order by g desc limit 1000;

```
Limit (cost=0.56..31.55 rows=1000 width=8) (actual time=0.023..0.378 rows=1000.00 loops=1)
  Buffers: shared hit=0
    → Index Only Scan Backward using g_index on students (cost=0.56..130065.23 rows=4197833 width=8) (actual time=0.022..0.253 rows=1000.00 loops=1)
      Index Cond: ((g > 10) AND (g < 20))
      Heap Fetches: 0
      Index Searches: 1
      Buffers: shared hit=8
Planning:
  Buffers: shared hit=20
Planning Time: 0.231 ms
Execution Time: 0.463 ms
(11 rows)
```